A SMART FRAUD DETECTION IN E-COMMERCE USING BLOCKCHAIN

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**A SMART FRAUD DETECTION IN E-COMMERCE USING BLOCKCHAIN** " has been carried out by the students **KAROLINA A**, **MADHUVADHANI S**, **MUBINA BEGUM M**, **RANJANI S** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

In today's increasingly digital world, fraudulent activities have become more sophisticated, posing significant challenges to organizations across industries such as banking, e-commerce, and insurance. Traditional rule-based systems often fail to detect new and evolving fraud patterns in real time. This paper presents a smart fraud detection system leveraging machine learning techniques to identify and mitigate fraudulent transactions more effectively. By utilizing supervised learning algorithms such as Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting, the system is trained on historical transaction data to recognize patterns indicative of fraud. The model is further enhanced through feature engineering, anomaly detection, and real-time data processing to improve accuracy and minimize false positives. Experimental results demonstrate high precision and recall rates, validating the model's ability to adapt to new fraud strategies. This intelligent approach significantly improves fraud detection efficiency and helps reduce financial losses, offering a scalable and robust solution for modern risk management.

Keywords: Blockchain, Fraud Detection, E-Commerce Security, Cryptography, Smart Contracts, Secure Transactions, Cybersecurity, Data Integrity.

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SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

OTP	-	One-Time Password
ML	-	Machine Learning
AI	-	Artificial Intelligence
DL	-	Deep Learning
API	-	Application Programming Interface
URL	-	Uniform Resource Locator
HTTP	-	Hypertext Transfer Protocol
HTTPS	-	Hypertext Transfer Protocol Secure
DB	-	Database
SQL	-	Structured Query Language
UI	-	User Interface
UX	-	User Experience
AES	-	Advanced Encryption Standard
RSA	-	Rivest–Shamir–Adleman (Encryption Algorithm)
SHA	-	Secure Hash Algorithm
PKI	-	Public Key Infrastructure
PoW	-	Proof of Work
PoS	-	Proof of Stake
DApp	-	Decentralized Application
P2P	-	Peer-to-Peer Network
KYC	-	Know Your Customer
GPU	-	Graphics Processing Unit
CPU	-	Central Processing Unit
JSON	-	JavaScript Object Notation

CHAPTER 1

INTRODUCTION

Credit card fraud in cybercrime refers to the use of technology with the help of internet to commit fraudulent activities to steal credit card information. This can include online purchases, data breaches, phishing scams, etc. This project involves a link protection scheme to maintain security while purchasing products through online platforms. This empowers the clients to perform the essential settings so as to manage bank transactions at their office or at homes through PC or laptop. The clients can access the bank's website to view their account details and perform transactions on record according to their requirements.With Internet Banking, the brick and mortar structure of traditional banking gets converted into a click and portal model, thus giving a virtual banking concept a real form. Today's banking is no more limited to branches. E-banking facilitates banking transactions by users round the clock globally.The primary aim of this software is to provide an enhanced design procedure, which visualizes future expansion and adaptation, which is vital for a core sector like banking. This requires the design to be expandable and modifiable thus a modular approach is utilized in building the software. The user having an account in bank can become a member of internet banking. All transactions are done online by transferring funds within the same bank.

Key considerations for secure online transactions include:

- Consider credit monitoring services
- Use virtual card numbers
- Always verify the authenticity of PAYMENT gateways

Figure. 1.1 Progress Diagram

1.1 OFFER SCAMS

Offer scams typically involve emails from someone claiming to need assistance moving money quickly. The sender includes just enough details to make the offer seem legitimate. However, the money transfer is inevitably delayed, leaving the victim responsible for various small payments to facilitate the transfer of funds.The widespread use of professional networking sites like LinkedIn has made unsolicited job offers common. Job seekers must become savvy at distinguishing legitimate offers from scams. If you decide to accept work, never cash suspicious checks without ensuring they are authentic. Ask the bank to place a hold on the funds until the check or money order is verified.

1.2 ONLINE SHOPPING SCAMS

Scammers use the latest technology to set up fake retailer websites that look like genuine online stores, using stolen logos and copied designs. Many of these websites offer popular brands of clothing, jewelry, or gadgets at low prices. Sometimes people may receive the item they paid for, but often they don't. A more recent version of the scam involves setting up a social media store,

Which usually disappears after a while to resurface again in another guise.The existing model of online shopping has significant vulnerabilities. When users place orders on platforms like Amazon, Flipkart, etc., they need to provide details like address, name, and phone number, and choose payment methods including credit card, UPI transactions, or cash on delivery. While cash on delivery is relatively safe, credit card transactions involve OTP verification that can be exploited by hackers.

When hackers obtain card details, they contact the cardholder using spam calls and collect the OTP through various pretexts:

- Posing as bank representatives claiming loan verification requirements
- Claiming the user has won a prize pool in a shopping mall
- Pretending to verify bank account details

Once the OTP is shared, money is debited for purchases made by hackers. By the time users notice the unauthorized transaction, it's often too late to take action.

1.3 OBJECTIVE OF THE PROJECT

The primary objective of this project is to enhance security in online transactions, which are payment methods involving electronic fund transfers over the internet. Online Transaction Processing (OLTP) should be secure and password protected. Three steps involved in online transactions are Registration, Placing an order, and Payment. The proposed system incorporates an enhanced security mechanism by implementing a secure login authentication process using two-factor authentication, thereby strengthening protection against unauthorized access.

Additionally, the system includes fraud detection and reporting features that identify suspicious transactions and provide immediate channels for reporting to cybercrime authorities. To further improve security, it promotes user awareness by sending transaction notifications, enabling users to make informed decisions before sharing OTPs. In case of fraudulent activity, the system offers legal recourse by providing direct security links that allow victims to quickly file complaints with cybercrime departments.It also ensures secure online generation of one-time passwords (OTPs), allowing reliable login procedures even in conditions of limited internet connectivity. Additionally, the system includes fraud detection and reporting features that identify suspicious transactions and provide immediate channels for reporting to cybercrime authorities. To further improve security, it promotes user awareness by sending transaction notifications, enabling users to make informed decisions before sharing OTPs.The primary objective of the study was to identify suspicious transactions during the payment process and prevent fraudulent activities before the transaction is completed. As online shopping continues to grow rapidly, the number of fraudulent transactions has also increased, making it essential to develop intelligent systems capable of detecting fraud in real time.In this system, transaction data such as purchase history, payment patterns, device information, and user behavior are continuously monitored and analyzed. A risk scoring model is used to assign a probability score to each transaction based on various parameters.

CHAPTER 2

LITERATURE SURVEY

2.1 REAL-TIME FRAUD DETECTION IN E-COMMERCE PLATFORMS IN 2025

M.Bhattacharya et al.This research focuses on developing a real-time fraud detection framework for e-commerce platforms using data analytics and machine learning techniques. The primary objective of the study was to identify suspicious transactions during the payment process and prevent fraudulent activities before the transaction is completed. As online shopping continues to grow rapidly, the number of fraudulent transactions has also increased, making it essential to develop intelligent systems capable of detecting fraud in real time.In this system, transaction data such as purchase history, payment patterns, device information, and user behavior are continuously monitored and analyzed. A risk scoring model is used to assign a probability score to each transaction based on various parameters. If the score exceeds a predefined threshold, the system flags the transaction as potentially fraudulent. Machine learning algorithms analyze historical transaction data to identify hidden patterns and anomalies associated with fraudulent behavior.the system learns complex fraud patterns that may not be easily detected by traditional rule- based systems. The results of this research show that real-time fraud detection systems can significantly reduce financial losses and enhance security in e- commerce platforms by enabling early detection of fraudulent transactions.The results of the study demonstrate that real-time fraud detection systems can significantly reduce financial losses and improve security in e-commerce environments. The proposed framework helps businesses identify suspicious activities quickly and take preventive actions such as transaction blocking or additional verification procedures.

2.2 DISTRIBUTED LEDGER TECHNOLOGY IN FINANCIAL FRAUD PREVENTION IN 2025

M. Crosby, P. Pattanayak, S. Verma et al.This research explores the use of Distributed Ledger Technology (DLT), commonly known as blockchain, to prevent financial fraud in digital transaction systems. The authors focus on the advantages of blockchain such as decentralization, immutability, and transparency, which can significantly enhance the security of financial systems. Traditional centralized databases are vulnerable to hacking and data manipulation, whereas blockchain ensures that transaction records cannot be altered once they are stored in the ledger.In the proposed system, every transaction is recorded as a block in a distributed ledger and validated through consensus mechanisms among network participants. Cryptographic hashing techniques are used to secure the blocks and link them together in a chain. This ensures that each transaction is authenticated and verified by multiple nodes within the network, making it extremely difficult for attackers to manipulate transaction data.This ensures that only valid transactions are added to the ledger. Each transaction is grouped into blocks, which are then linked together using cryptographic hash functions to form a secure chain of blocks. The study concludes that blockchain technology can serve as a powerful solution for preventing fraud in financial systems. By eliminating single points of failure and enabling secure peer-to-peer transactions, distributed ledger technology can improve trust, transparency, and reliability in digital payment environments. Artificial intelligence techniques are applied to analyze transaction patterns and detect anomalies that may indicate fraudulent activities.The study concludes that decentralized blockchain-based systems can significantly improve security and transparency in e-commerce platforms. By eliminating intermediaries and ensuring immutable transaction records, blockchain technology provides a reliable solution for secure online commerce. In the proposed framework, transactions are recorded on a distributed ledger using blockchain technology.

2.3 HYBRID BLOCKCHAIN AND AI-BASED FRAUD DETECTION SYSTEM IN 2025

L.Chen, K. Salah, M. Habib Ur Rehman et al.This research proposes a hybrid fraud detection system that combines blockchain technology with artificial intelligence to enhance the security of financial transactions. The main objective of the study was to create a secure and intelligent framework capable of detecting fraudulent transactions whilemaintaining transparency and data integrity. The integration of blockchain and AI helps address limitations found in traditional fraud detection systems.In the proposed architecture, blockchain is used to securely record transaction data using decentralized ledgers and cryptographic security mechanisms. Smart contracts are implemented to automate transaction verification processes and enforce security rules within the blockchain network. Artificial intelligence techniques are applied to analyze transaction patterns and detect anomalies that may indicate fraudulent activities.The study concludes that decentralized blockchain-based systems can significantly improve security and transparency in e-commerce platforms. By eliminating intermediaries and ensuring immutable transaction records, blockchain technology provides a reliable solution for secure online commerce. In the proposed framework, transactions are recorded on a distributed ledger using blockchain technology. Encryption techniques and cryptographic hashing are used to secure transaction data and protect sensitive user information. Ethereum-based smart contracts are implemented to automatically verify transactions and enforce business rules within the system. The study concludes that decentralized blockchain-based systems can significantly improve security and transparency in e-commerce platforms.

2.4 BEHAVIORAL ANALYSIS FOR DETECTING ONLINE PAYMENT FRAUD IN 2024

R.Davis, V. J. Hodge, J. Austin et al.This research investigates the use of behavioral analysis techniques to detect fraudulent activities in online payment systems. The authors focus on analyzing user behavior patterns such as transaction frequency, purchase locations, device usage, and login activities. By monitoring these behavioral attributes, the system can identify deviations from normal user behavior and detect potential fraud attempts. In the proposed architecture, blockchain is used to securely record transaction data using decentralized ledgers and cryptographic security mechanisms. Smart contracts are implemented to automate transaction verification processes and enforce security rules within the blockchain network. Artificial intelligence techniques are applied to analyze transaction patterns and detect anomalies that may indicate fraudulent activities. In this approach, anomaly detection techniques and statistical modeling methods are used to analyze user behavior data. Machine learning algorithms are applied to build predictive models that distinguish between legitimate and fraudulent transactions. The system continuously learns from new transaction data, enabling it to adapt to evolving fraud patterns over time.The results of the research demonstrate that behavioral analysis can significantly improve fraud detection accuracy in online payment systems. By identifying unusual transaction patterns and behavioral deviations, the system can alert users or financial institutions before significant financial damage occurs. In the proposed framework, transactions are recorded on a distributed ledger using blockchain technology. Encryption techniques and cryptographic hashing are used to secure transaction data and protect sensitive user information. Ethereum-based smart contracts are implemented to automatically verify transactions and enforce business rules within the system. The study concludes that decentralized blockchain-based systems can significantly improve security and transparency in e-commerce platforms.

2.5 DECENTRALIZED E-COMMERCE SECURITY USING BLOCKCHAIN IN 2024

M.Elangovan, Z. Zheng, S. Xie et al.This research explores the use of blockchain technology to enhance security in e-commerce platforms by implementing a decentralized transaction system. The authors emphasize that traditional e-commerce platforms rely heavily on centralized servers, which are vulnerable to hacking, data breaches, and fraudulent manipulation. Blockchain technology offers a decentralized solution that can improve transparency and trust between buyers and sellers. In the proposed framework, transactions are recorded on a distributed ledger using blockchain technology. Encryption techniques and cryptographic hashing are used to secure transaction data and protect sensitive user information. Ethereum-based smart contracts are implemented to automatically verify transactions and enforce business rules within the system. The study concludes that decentralized blockchain-based systems can significantly improve security and transparency in e-commerce platforms. By eliminating intermediaries and ensuring immutable transaction records, blockchain technology provides a reliable solution for secure online commerce. In this approach, anomaly detection techniques and statistical modeling methods are used to analyze user behavior data. Machine learning algorithms are applied to build predictive models that distinguish between legitimate and fraudulent transactions. The system continuously learns from new transaction data, enabling it to adapt to evolving fraud patterns over time.The results of the research demonstrate that behavioral analysis can significantly improve fraud detection accuracy in online payment systems. By identifying unusual transaction patterns and behavioral deviations, the system can alert users or financial institutions before significant financial damage occurs.

2.6 BLOCKCHAIN-BASED SECURE E-COMMERCE TRANSACTION SYSTEM IN 2024

P. Gupta, Satoshi Nakamoto et al.This research discusses the fundamental concept of blockchain technology for secure digital transactions. The author introduces blockchain as a decentralized system where transactions are verified and recorded across multiple nodes in a distributed network. This eliminates the need for centralized authorities and reduces the risk of fraud and data manipulation.In the proposed system, cryptographic algorithms are used to secure transaction data, while a distributed ledger ensures transparency and traceability. Each transaction is grouped into blocks, which are linked together using cryptographic hashes to form a secure blockchain. Once recorded, these transactions cannot be altered or deleted, ensuring data integrity.The study highlights that blockchain technology can significantly enhance the security of e-commerce transactions by providing a transparent, tamper-proof, and decentralized transaction environment.The research highlights that blockchain technology provides a secure and transparent framework for conducting financial transactions in e- commerce systems. By using distributed ledgers and cryptographic verification mechanisms, blockchain can significantly improve the reliability and security of online transaction platforms. In this approach, anomaly detection techniques and statistical modeling methods are used to analyze user behavior data. Machine learning algorithms are applied to build predictive models that distinguish between legitimate and fraudulent transactions. The system continuously learns from new transaction data, enabling it to adapt to evolving fraud patterns over time.The results of the research demonstrate that behavioral analysis can significantly improve fraud detection accuracy in online payment systems. By identifying unusual transaction patterns and behavioral deviations, the system can alert users or financial institutions before significant financial damage occurs.

2.7 CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING IN 2023

J. Lee, Andrea Dal Pozzolo , Gianluca Bontempi et al.This research focuses on the detection of credit card fraud using machine learning techniques. With the rapid growth of online payment systems, credit card fraud has become a major concern for financial institutions and e- commerce platforms. Traditional fraud detection systems rely mainly on manual verification and rule-based techniques, which are often unable to detect sophisticated fraud patterns. Therefore, the authors propose the use of machine learning algorithms to automatically identify fraudulent transactions by analyzing large volumes of financial data. In the proposed system, historical credit card transaction datasets are used to train machine learning models. These datasets contain both legitimate and fraudulent transaction records. Before applying machine learning algorithms, data preprocessing techniques are used to clean the dataset and handle issues such as missing values and class imbalance. Feature engineering techniques are also applied to extract important transaction attributes such as transaction amount, time of transaction, and frequency of purchases. The researchers applied machine learning techniques such as Logistic Regression, Random Forest, and data mining algorithms to classify transactions as legitimate or fraudulent. Historical transaction data was used to train the models, allowing them to learn behavioral patterns associated with fraud. Data preprocessing techniques were also used to handle imbalanced datasets and improve model accuracy. These contracts reduce the need for intermediaries and ensure that transactions are executed only when predefined conditions are satisfied. The research concludes that blockchain technology has the potential to transform e- commerce platforms by improving transparency, security, and trust among participants. The research concludes that blockchain technology has the potential to transform e-commerce platforms by providing secure, transparent, and decentralized transaction infrastructures.

2.8 A SURVEY ON BLOCKCHAIN TECHNOLOGY FOR E-COMMERCE APPLICATIONS IN 2023

Olivier Caelen et al.This research provides a comprehensive survey of blockchain technology and its potential applications in e-commerce systems. The rapid expansion of online marketplaces has increased the demand for secure and transparent transaction systems. Traditional e-commerce platforms rely on centralized architectures that require trust in third-party intermediaries such as payment gateways and financial institutions. However, these centralized systems are vulnerable to cyberattacks, fraud, and data manipulation. The author discusses how blockchain technology can address these challenges by providing a decentralized and secure infrastructure for digital transactions. Blockchain uses distributed ledger technology to store transaction records across multiple nodes in a network. Each transaction is verified through cryptographic algorithms and consensus mechanisms before being added to the blockchain. This ensures that transaction data remains immutable and tamper- proof. The survey also explores the role of smart contracts in automating e- commerce processes. Smart contracts are self-executing programs stored on the blockchain that automatically enforce agreements between buyers and sellers. These contracts reduce the need for intermediaries and ensure that transactions are executed only when predefined conditions are satisfied. The research concludes that blockchain technology has the potential to transform e- commerce platforms by improving transparency, security, and trust among participants. The research concludes that blockchain technology has the potential to transform e-commerce platforms by providing secure, transparent, and decentralized transaction infrastructures.The survey also explores the role of smart contracts in automating e- commerce processes. Smart contracts are self-executing programs stored on the blockchain that automatically enforce agreements between buyers and sellers.

2.9 FRAUD DETECTION IN ONLINE TRANSACTIONS USING MACHINE LEARNING IN 2023

M. Swan, Bhattacharyya Et AL et al.This research examines the use of machine learning techniques to detect fraudulent activities in online financial transactions. The authors emphasize that fraud detection is a complex problem due to the constantly evolving nature of fraudulent activities. Fraudsters continuously develop new techniques to bypass traditional security mechanisms, making it necessary to develop intelligent systems that can adapt to changing fraud patterns. In the proposed approach, transaction data is collected and analyzed using machine learning algorithms such as Decision Trees and Neural Networks. These algorithms are capable of learning complex relationships between transaction attributes and identifying patterns associated with fraudulent behavior. The system uses historical transaction data to train predictive models that can classify new transactions based on their likelihood of being fraudulent.Behavioral analysis techniques are also incorporated into the system to analyze user behavior patterns such as purchase frequency, transaction locations, and payment methods. By comparing new transactions with historical user behavior profiles, the system can detect abnormal activities that may indicate fraud. The research findings demonstrate that machine learning-based fraud detection systems can significantly improve the accuracy and efficiency of fraud prevention mechanisms in online transaction platforms.The research findings indicate that machine learning-based fraud detection systems can significantly enhance the security of online transaction platforms. These systems provide automated monitoring capabilities and help financial institutions detect suspicious transactions in real time.The survey also explores the role of smart contracts in automating e- commerce processes. Smart contracts are self-executing programs stored on the blockchain that automatically enforce agreements between buyers and sellers.

2.10 SMART CONTRACTS FOR SECURE TRANSACTION PROCESSING IN 2022

V. Zangking et al.This research introduces the concept of smart contracts as a secure method for automating digital transactions. Smart contracts are self-executing programs stored on a blockchain that automatically enforce agreements between parties without requiring intermediaries. The primary objective of the study was to demonstrate how smart contracts can improve security and efficiency in financial transactions. In the proposed system, smart contracts are developed using blockchain platforms such as Ethereum. These contracts automatically execute predefined conditions when certain criteria are met. The use of cryptographic verification ensures that contract rules cannot be altered or manipulated once deployed on the blockchain network. The study concludes that smart contracts provide a reliable and secure method for processing transactions in decentralized systems. By automating transaction verification and eliminating intermediaries, smart contracts can reduce fraud risks and improve efficiency in digital financial systems. The research findings demonstrate that machine learning-based fraud detection systems can significantly improve the accuracy and efficiency of fraud prevention mechanisms in online transaction platforms.The research findings indicate that machine learning-based fraud detection systems can significantly enhance the security of online transaction platforms. These systems provide automated monitoring capabilities and help financial institutions detect suspicious transactions in real time.

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

The existing system of online shopping platforms such as Amazon and Flipkart allows users to purchase products through digital platforms by creating an account and placing orders through the website or mobile application. In this system, users must provide personal details such as name, address, phone number, and email address while creating their accounts. Once the account is created, users can browse products, add items to their cart, and proceed with the checkout process. The system offers different payment options such as credit card payment, UPI transactions, and cash on delivery. Among these methods, cash on delivery is considered relatively safer since the payment is made only when the product is delivered to the customer.

However, when users choose online payment methods such as credit card transactions or digital payments, the system usually relies on an OTP (One-Time Password) verification mechanism to authenticate the transaction. In this process, after entering card details such as credit card number, billing address, city, and pin code, an OTP is sent to the registered mobile number of the cardholder. The user must enter the OTP within a limited time to complete the transaction successfully. Although OTP-based verification is intended to provide an additional layer of security, it has become one of the major targets for cybercriminals who attempt to exploit users through social engineering techniques.In many fraud scenarios, attackers obtain partial user data through phishing attacks, data leaks, or malicious websites. Once they gain access to certain information, they contact the cardholder through spam calls or fake messages pretending to be representatives from banks, shopping platforms, or customer support services. For example, attackers may claim that the user’s bank account requires verification or that the user has won a prize or reward

related to a recent purchase. Once the OTP is shared, the attacker can complete the transaction and withdraw money from the victim's account without authorization.Another major issue in the existing system is that once the fraudulent transaction is completed, the user often receives a notification only after the amount has already been deducted from the account. At that stage, it becomes difficult for the user to immediately stop the transaction or recover the lost money.

Although banks may offer complaint mechanisms or fraud reporting options, the response time is often slow, and the chances of recovering the funds may be limited. This delay in fraud detection creates a major security concern in online payment systems.The existing e-commerce model typically involves several basic processes such as account creation, user authentication, product selection, cart management, and payment processing. During account creation, users are required to enter personal information including their name, email address, phone number, password, and password confirmation. Once registered, users can sign in using their email address and password. After logging in, users can browse products available on the platform, add items to their shopping cart, and proceed with the checkout process. When placing an order, users must enter their bank card details, billing address, shipping address, and payment method to complete the transaction.Despite providing convenient shopping experiences, the existing system has several limitations in terms of transaction security and fraud prevention. OTP verification alone cannot guarantee complete protection against fraud because attackers can manipulate users into sharing OTP codes through deception or social engineering techniques.

In addition, the system lacks advanced mechanisms for monitoring suspicious transaction activities or detecting fraudulent behavior in real time. Therefore, the existing online shopping model faces significant challenges in ensuring secure financial transactions. The absence of strong fraud detection mechanisms, combined with the increasing

Disadvantages of Existing System

Online payment transactions currently suffer from limited security, as OTP verification alone is not sufficient to prevent fraud. In many cases, fraudulent transactions are detected only after the amount has already been debited, leaving users with limited control once the payment process is completed. Additionally, there is a lack of transparency in transaction verification processes, and no decentralized verification mechanism is in place to enhance security. These weaknesses make the system highly vulnerable to phishing attacks and social engineering scams, further increasing the risk of financial loss for users.

3.2 BLOCK DIAGRAM OF EXISTING SYSTEM

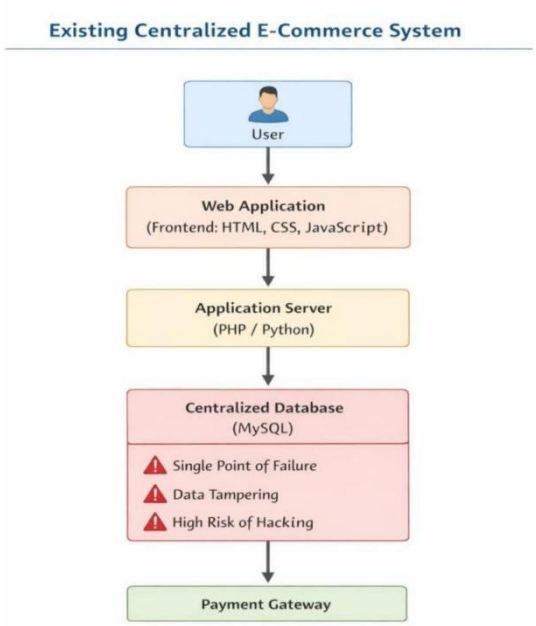

Figure. 3.1. Existing Diagram

CHAPTER 4

PROBLEMS IDENTIFIED

Despite the rapid growth of e-commerce platforms and digital payment systems, existing online shopping environments still face several security challenges. These issues increase the risk of financial fraud, data theft, and unauthorized transactions.One of the major problems in the existing system is OTP-based fraud through social engineering attacks. Hackers often contact users through fake calls, messages, or emails pretending to be bank officials or customer support representatives. They manipulate users into sharing the OTP sent to their registered mobile number. Once the OTP is revealed, attackers can complete unauthorized transactions and withdraw money from the user's account. The following problems have been identified in the current e- commerce transaction systems:

4.1 REAL TIME PROBLEMS

4.1.1 Vulnerability of OTP-Based Authentication

Most online payment systems rely on OTP (One-Time Password) verification for transaction authentication. However, attackers often exploit social engineering techniques such as phishing calls, fake messages, and fraudulent emails to trick users into revealing their OTP. Once the OTP is shared, hackers can easily complete unauthorized transactions and withdraw money from the user's account.

4.1.2 Lack of Real-Time Fraud Detection

Existing systems primarily focus on verifying user credentials rather than analyzing transaction behavior. They do not actively monitor suspicious transaction patterns or unusual activities during the payment process. As a

result, fraudulent transactions are usually detected only after the money has been transferred, making recovery difficult.

4.1.3 Centralized Data Storage Risks

Most e-commerce platforms store user data and transaction records in centralized databases. These centralized systems create a single point of failure, making them vulnerable to hacking, data breaches, and unauthorized access. If attackers gain access to the database, sensitive user information such as card details and personal data may be compromised.

4.1.4 Lack of Transparency in Transactions

In traditional e-commerce systems, users cannot independently verify whether their transaction data has been altered or manipulated. The lack of transparency reduces trust between users, payment gateways, and online platforms, especially when disputes arise regarding failed or fraudulent transactions.

4.1.5 Weak Fraud Prevention Mechanisms

Current systems mainly depend on password authentication and OTP verification, which are not always sufficient to prevent fraud. Advanced attackers can bypass these security layers through phishing attacks, malware, or social engineering techniques. This shows the need for stronger and more secure fraud prevention mechanisms.

4.1.6 Delayed Fraud Response

When a fraudulent transaction occurs, users usually receive notification only after the payment has been completed. At that stage, reversing the transaction

becomes complex and time-consuming. The absence of immediate fraud detection and prevention systems increases the financial loss for users.

4.1.7 Risk of Data Tampering

Since transaction records are stored in centralized servers, there is a possibility that attackers or unauthorized insiders may alter or manipulate transaction information. This compromises the integrity and reliability of the transaction data.

4.1.8 Lack of Secure Transaction Verification

Most existing systems do not use advanced cryptographic verification methods for securing transactions. Without strong encryption and decentralized validation mechanisms, sensitive financial data remains exposed to cyber threats.

4.1.9 Limited User Trust in Online Payments

Frequent reports of online fraud and payment scams reduce user confidence in digital transactions. Many users hesitate to use online payment systems due to concerns about security, privacy, and financial safety.

4.1.10 Absence of Decentralized Security Mechanisms

Traditional e-commerce platforms depend entirely on centralized authorities for transaction verification and security. The absence of decentralized technologies such as blockchain limits the ability to create tamper-proof and transparent transaction records.

CHAPTER 5

PROPOSED SYSTEM

5.1 INTRODUCTION

The proposed system introduces a secure blockchain-based fraud detection system for e-commerce platforms to overcome the limitations of traditional online transaction systems. The primary objective of this system is to enhance the security, transparency, and reliability of online payment transactions while protecting users from fraudulent activities. Unlike the existing centralized systems, the proposed solution uses blockchain technology and cryptographic techniques to ensure that transaction data is stored in a decentralized and tamper-proof ledger.In this system, when a user places an order on an e-commerce platform, the transaction details are securely recorded in the blockchain network. Each transaction is verified using cryptographic hashing and distributed across multiple nodes, which prevents unauthorized modifications or data manipulation. This decentralized verification mechanism ensures that every transaction is transparent and cannot be altered once it is recorded in the blockchain ledger.The system also incorporates secure authentication and transaction validation mechanisms to protect users from OTP-based fraud attacks. Instead of relying solely on traditional authentication methods, the proposed system strengthens security by ensuring that transaction records are verified and validated within the blockchain network before completion. This helps prevent unauthorized transactions and provides a higher level of trust between users, merchants, and payment systems.Additionally, the proposed platform maintains an immutable record of all transactions, allowing administrators and users to track and verify transaction history whenever required.

Advantages of Proposed System

The proposed system enhances transaction transparency by enabling both users and administrators to verify transaction records without the risk of data manipulation. It effectively prevents OTP-based fraud and unauthorized transactions through secure cryptographic validation mechanisms. By leveraging blockchain technology, the system ensures data integrity, meaning that once a transaction is recorded, it cannot be modified or deleted. Additionally, it reduces the risk of data breaches and cyberattacks by eliminating the single point of failure commonly found in centralized systems. This approach builds greater trust among users, merchants, and payment platforms by maintaining transparent and verifiable records. Overall, the platform provides secure and reliable online payment processing, significantly improving the safety of e-commerce transactions.

5.2 BLOCK DIAGRAM OF PROPOSED SYSTEM

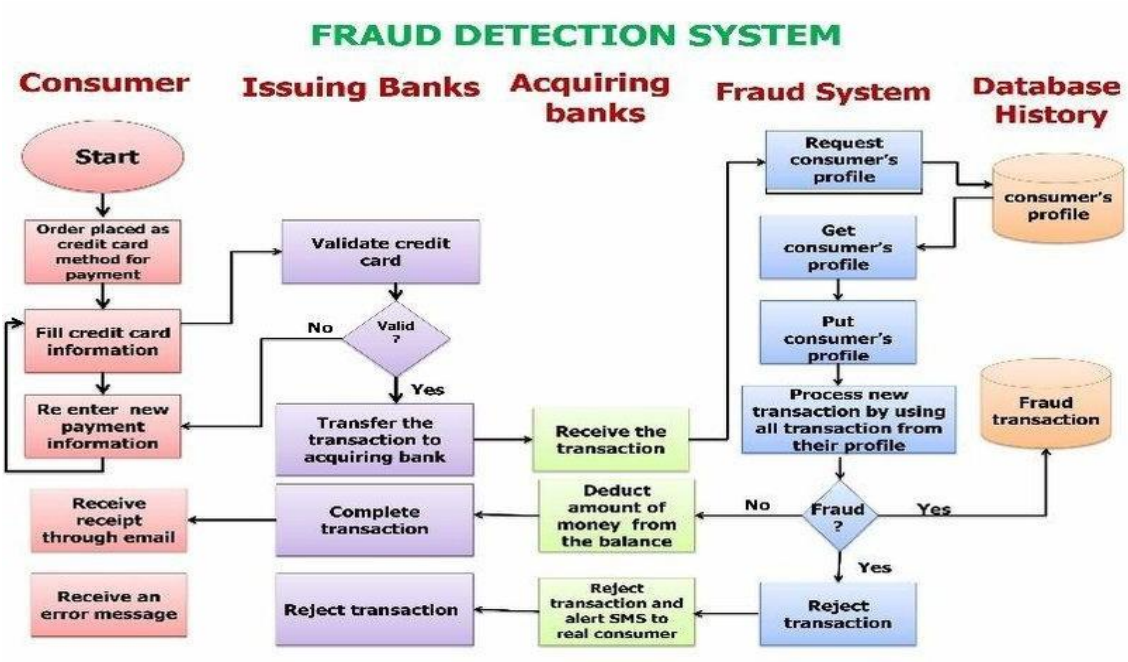

Figure. 5.1. Proposed Diagram

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel® Core™ i5 processor 4300M at 2.60 GHz or 2.59
RAM	Minimum 8 GB
Storage	50 GB or above
Display	Standard Monitor

6.2 SOFTWARE REQUIREMENTS

Component	Specification
Operating System	Windows / Linux /macOS
Server Side	PHP (64-bit) or (32-bit)
Client Side	HTML,CSS,Bootstrap
Database	MySql
Server	Xamp

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

- User Module – Registration, login, purchase
- Seller Module – Product listing, order management
- Admin Module – Monitoring & fraud reports
- Blockchain Module – Transaction recording
- Fraud Detection Module – Risk scoring system
- Alert & Notification Module – Real-time alert

7.2 MODULES DESCRIPTION

7.2.1 USER MODULE

The User Module is responsible for managing the activities performed by customers on the e-commerce platform. This module allows users to create a new account by providing basic details such as name, email address, phone number, and password. Once registered, users can securely log in to the system using their credentials. The authentication process verifies the entered details with the stored database records to ensure secure access to the platform.After successful login, users can browse different products available on the platform. The system provides detailed information about each product including product name, description, price, and seller details. Users can select the desired products, add them to the cart, and proceed with the purchasing process.

The module also manages order placement and transaction initiation. When a user confirms a purchase, the system records the order details and initiates the payment transaction. This module ensures that all user interactions with the platform are secure, reliable, and easy to use.Finally, when the user confirms the purchase, the system initiates the payment process and records the

transaction details. The purchase information is forwarded to the payment system and blockchain module for verification and secure transaction recording. This module ensures a smooth and secure shopping process for users while maintaining accurate records of their transactions.

7.2.2 SELLER MODULE

The Seller Module allows vendors or merchants to register on the platform and manage their products effectively. Sellers can create accounts by entering business information such as seller name, contact details, and product categories. After registration, sellers can log in to their accounts and access the seller dashboard.Through this module, sellers can add new products to the platform by entering product details such as product name, description, price, quantity, and images. The system stores this information in the database and displays it to users who are browsing the platform.

Additionally, the Seller Module helps sellers manage incoming orders placed by customers. Sellers can view order details, update order status, and manage product availability. This module ensures that sellers can efficiently maintain their product listings and fulfill customer orders without delays.Another important feature of the Seller Module is order management. Sellers can view orders placed by customers and track their order status through the dashboard. The system displays order details such as customer information, product details, payment status, and delivery information. This helps sellers efficiently process orders and manage product deliveries.

7.2.3 ADMIN MODULE

The Admin Module provides administrative control over the entire system and ensures smooth operation of the platform. Administrators are responsible for managing users, sellers, products, and transaction activities. This module includes a secure login interface that allows only authorized

administrators to access the system management panel.Through the admin dashboard, administrators can monitor all user registrations and verify seller accounts before allowing them to list products on the platform. This verification process ensures that only legitimate sellers are allowed to operate within the system. By maintaining proper control over registrations, the administrator helps maintain trust and reliability within the platform.

The module also allows administrators tomanage product listings and monitor seller activities. If any product violates platform rules or contains incorrect information, the administrator has the authority to modify or remove the listing. This helps maintain the quality and authenticity of the products displayed on the platform.

Another important function of the Admin Module is monitoring transaction activities and identifying suspicious patterns. The system provides access to transaction records, which can be analyzed to detect unusual activities such as repeated failed payments, abnormal transaction amounts, or unauthorized access attempts.In addition, the module generates reports related to fraud detection, transaction history, and system activity logs. These reports help administrators analyze system performance and improve security measures. By continuously monitoring the platform, the admin ensures the safety and reliability of the entire system.

In addition, the module generates reports related to fraud detection, transaction history, and system activity logs. These reports help administrators analyze system performance and improve security measures. By continuously monitoring the platform, the admin ensures the safety and reliability of the entire system.

7.2.4 BLOCKCHAIN MODULE

The Blockchain Module is responsible for maintaining secure and transparent records of all financial transactions occurring within the system. Blockchain technology provides a decentralized ledger where transaction data is stored in blocks that are linked together using cryptographic techniques. This structure ensures that once a transaction is recorded, it cannot be altered or tampered with.When a user completes a purchase, the transaction details are securely transmitted to the blockchain module. These details include information such as transaction ID, user ID, product details, payment amount, and timestamp. The system then creates a block containing this information and generates a cryptographic hash to secure the block.

Each newly created block is connected to the previous block through a hash value, forming a continuous chain of blocks. This chain structure ensures data integrity and prevents unauthorized modification of transaction records. Any attempt to alter a block would change its hash value and break the chain, making the tampering easily detectable.Another advantage of blockchain technology is decentralization. Instead of storing data in a single centralized server, the blockchain network distributes transaction records across multiple nodes. This reduces the risk of data loss, hacking, or system failure caused by a single point of failure.

By using blockchain technology, the system improves transparency, reliability, and security of online transactions. Users and administrators can verify transaction records whenever necessary, making the system more trustworthy and resistant to fraudulent activities.

7.2.5 FRAUD DETECTION MODULE

The Fraud Detection Module plays a critical role in protecting the e-commerce platform from fraudulent transactions. Online payment systems are often targeted by cybercriminals who attempt to perform unauthorized transactions using stolen card details or OTP scams. This module is designed to detect such suspicious activities and prevent financial losses.

The module works by analyzing transaction data and identifying unusual patterns or behaviors that may indicate fraud. Several parameters are considered during this analysis, including transaction amount, frequency of transactions, location information, and user activity patterns. These factors help the system determine whether a transaction is legitimate or suspicious.

Based on the analysis results, the system assigns a risk score to each transaction. Transactions with normal behavior receive a low risk score and are processed normally. However, transactions that show unusual patterns receive a higher risk score and may be flagged for further verification.When a transaction is identified as suspicious, the system generates alerts and temporarily pauses the transaction process. The flagged transaction can then be reviewed by the administrator or verified through additional security checks. This helps prevent fraudulent transactions before the payment is completed.

7.2.6 ALERT AND NOTIFICATION MODULE

The Alert and Notification Module is responsible for keeping users and administrators informed about important system activities. In an online transaction environment, timely communication is essential to ensure transparency and maintain user trust. This module generates notifications for various events occurring within the system.Whenever a user successfully completes a transaction or places an order, the system automatically sends a

confirmation notification. This notification may include order details, payment status, and expected delivery information. These updates help users stay informed about their purchases and improve their overall experience.

The module also plays an important role in fraud prevention. If the fraud detection module identifies a suspicious transaction, an alert is immediately sent to the administrator for further investigation. This quick notification enables administrators to take prompt action and prevent potential financial losses.In addition to fraud alerts, the module also notifies users about system updates, order status changes, and other important messages. These notifications ensure that users remain aware of all relevant activities related to their accounts and transactions.

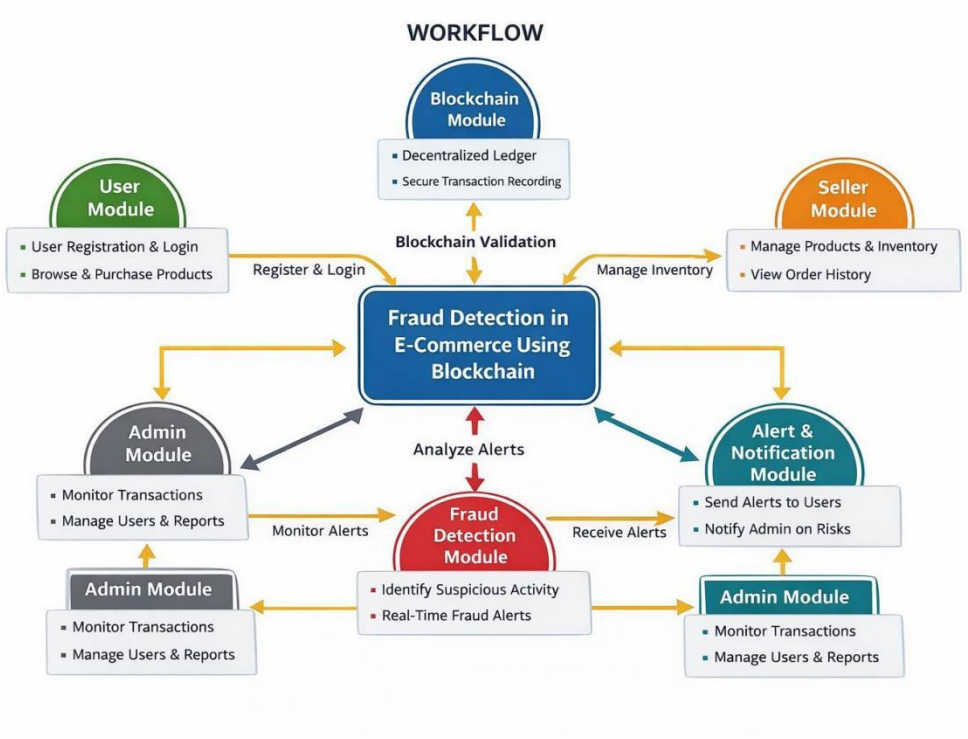

Figure. 7.1. Workflow Diagram

CHAPTER 8

SYSTEM TESTING

System testing is an important stage in software development that ensures the system works correctly and meets the required specifications. The testing process verifies whether all modules of the system operate properly and interact effectively with each other. Different testing techniques are used to examine the functionality, performance, security, and usability of the system. These tests help identify errors, validate system behavior, and ensure that the application performs reliably in real-world conditions.

8.1. UNIT TESTING

Unit testing is a software testing technique that focuses on verifying the functionality of individual components or modules independently. The main objective of unit testing is to ensure that each module performs its intended function correctly without depending on other modules. By testing smaller parts of the system separately, developers can easily identify coding errors and logical issues at an early stage of development. This helps improve the reliability and stability of the software system. In the proposed system, unit testing is performed on various modules such as the user authentication module, product management module, transaction module, and blockchain verification module. Each module is tested individually to confirm that the functions within the module produce the expected results. This testing ensures that every component of the system works correctly before it is integrated with other modules.

During unit testing, different input values are provided to each function, and the outputs are carefully examined. For example, the user registration module is tested to ensure that valid user details are correctly stored in the database and invalid inputs are rejected. Similarly, the login module is

tested to verify that the authentication process correctly validates user credentials.Unit testing also verifies that database operations such as inserting, updating, and retrieving records work properly. By checking these operations independently, developers ensure that data is stored and retrieved accurately without errors.

Overall, unit testing plays a vital role in improving the quality of the system. By identifying and fixing errors at the module level, it prevents issues during later stages such as integration testing and system testing.

8.2. INTEGRATION TESTING

Integration testing is performed to verify that different modules of the system work correctly when combined together. After individual modules are tested using unit testing, integration testing ensures that the modules communicate effectively and exchange data without errors. The main goal of integration testing is to detect problems that may occur when different components interact with each other.In the proposed system, integration testing is conducted after combining modules such as user registration, product management, transaction processing, blockchain verification, and fraud detection. These modules must work together smoothly to ensure that the entire transaction process functions correctly.

During integration testing, the output generated by one module becomes the input for another module. For example, when a user places an order, the system transfers the transaction details to the blockchain module for secure recording. Integration testing verifies that this data transfer occurs correctly without any loss or corruption of information.

The testing process also checks whether the fraud detection module receives transaction data correctly and analyzes it for suspicious activities. If any transaction appears suspicious, the system should generate an alert and

notify the administrator.Through integration testing, developers confirm that the entire system workflow operates smoothly. This testing ensures that all modules communicate properly and function together as a complete application.

8.3. SYSTEM TESTING

System testing evaluates the entire application as a complete system. The main objective of system testing is to verify whether the developed system satisfies all functional and non-functional requirements. Unlike unit testing and integration testing, which focus on individual components, system testing examines the behavior of the full system in a real operating environment.In this project, system testing is performed after all modules have been successfully integrated. The system is tested using real-world scenarios where users register accounts, browse products, and perform transactions. This process helps confirm that the application behaves correctly when all components operate together.

During system testing, the transaction workflow is tested from start to finish. The system processes user input, verifies payment details, records the transaction on the blockchain network, and generates confirmation notifications. Each stage of the process is observed to ensure that it produces accurate results.System testing also verifies the correctness of data processing and transaction recording. The system ensures that every transaction is securely stored and cannot be modified after recording. This confirms the reliability of the blockchain-based transaction system.

Overall, system testing ensures that the entire application operates efficiently and meets the expected requirements. It validates the system's functionality and confirms that the platform works properly for real users.

8.4. PERFORMANCE TESTING

Performance testing evaluates how efficiently the system performs under different operating conditions. The main objective of this testing is to measure system speed, responsiveness, and stability when handling user requests and transaction processes. Performance testing ensures that the system maintains acceptable performance levels even during heavy usage.In this project, performance testing focuses on evaluating the efficiency of the transaction processing system and blockchain verification mechanism. The system is tested with multiple transactions to measure how quickly the application processes data and records transactions in the blockchain network.

During this testing phase, several performance metrics are analyzed, including response time, transaction processing speed, and system resource usage. Response time measures how quickly the system reacts to user requests, while processing speed determines how fast transactions are completed.Stress testing is also performed to examine how the system behaves under high workload conditions. Multiple transactions are executed simultaneously to determine whether the system can handle large volumes of data without crashing or slowing down significantly.

The results of performance testing help identify areas where system performance can be improved. By optimizing the system for speed and efficiency, developers ensure that users experience smooth and reliable operation.

8.5. SECURITY TESTING

Security testing is conducted to ensure that the system protects sensitive information and prevents unauthorized access. The main objective of this testing is to identify potential vulnerabilities that could allow attackers to access or manipulate system data. Security testing is especially important for systems that handle financial transactions and user information.In the proposed system,

security testing focuses on protecting user account details, transaction information, and blockchain records. The system verifies that only authorized users can access their accounts and perform transactions. Authentication mechanisms such as secure login validation are tested to prevent unauthorized access.

Another important aspect of security testing is verifying that the blockchain transaction records cannot be modified or deleted by unauthorized users. The system ensures that all transaction data is stored securely and protected using cryptographic techniques.Input validation is also tested to prevent malicious attacks such as SQL injection and data manipulation. All user inputs are validated before processing to ensure that only valid data is accepted by the system.Through these security measures, the system maintains data integrity, protects user privacy, and prevents cyber threats.

8.6. USABILITY TESTING

Usability testing is an essential phase in the development of the proposed fraud detection system, as it evaluates how easily users can interact with the application and perform tasks efficiently. The main objective of usability testing is to ensure that the system is user-friendly, intuitive, and accessible to both technical and non-technical users. In this project, usability testing focuses on analyzing the ease of navigation, clarity of interface design, responsiveness of the system, and overall user satisfaction during online transactions.

The testing process involved a group of users who interacted with the system by performing common tasks such as user registration, login authentication, product selection, and secure payment processing. Special attention was given to the fraud detection features, including transaction alerts, verification steps, and blockchain-based confirmation mechanisms. Users were asked to complete these tasks without prior training to evaluate the system's learnability and ease of use.

The results of usability testing indicated that the system provides a smooth and efficient user experience. The interface was designed with simple navigation menus, clear instructions, and well-structured layouts, enabling users to easily understand and operate the system. The integration of secure transaction steps, such as enhanced verification and alert notifications, was found to be effective without causing confusion or delay in the payment process.

Feedback collected from users highlighted that the system successfully balances security and usability. While advanced technologies like blockchain and cryptographic validation operate in the background, the front-end remains simple and user-friendly. Users appreciated the transparency provided by transaction records and the immediate alerts for suspicious activities, which increased their confidence in using the platform.

CHAPTER 9

RESULTS AND DISCUSSION

In the existing online shopping systems, payment verification mainly depends on methods such as OTP authentication, card verification, or other traditional security mechanisms. Although these methods provide an additional level of protection, they are still vulnerable to different types of cyber attacks. Hackers often use social engineering techniques such as phishing messages, fake customer care calls, and spam emails to collect OTPs from users. Once the OTP is shared with attackers, unauthorized transactions can be performed easily, which results in financial loss for users.

The analysis of the existing system clearly indicates that OTP-based verification alone is not sufficient to ensure secure financial transactions. Attackers can manipulate users into revealing sensitive information such as OTPs, card numbers, or personal details. Moreover, once a transaction is completed, it becomes difficult to track or reverse fraudulent activities because the transaction data is stored in centralized databases that may be vulnerable to tampering or unauthorized access.

To overcome these issues, the proposed system introduces blockchain technology to enhance the security of transaction records. Blockchain is a decentralized digital ledger that stores transaction data in blocks connected through cryptographic hashes. Each block contains information about the transaction, a timestamp, and the hash value of the previous block. This structure ensures that once a transaction is recorded in the blockchain, it cannot be modified or deleted without affecting the entire chain.

During the implementation of the proposed system, all transaction activities performed by users are recorded in the blockchain ledger. When a user places an order and completes a payment, the transaction information is converted into a block and added to the blockchain network. Each block contains a unique hash value that ensures the integrity and authenticity of the stored data. This mechanism prevents unauthorized modifications and ensures that transaction records remain secure and transparent.

The usability evaluation of the system shows that the application provides a simple and user-friendly interface for customers. Users can easily register, browse products, add items to the shopping cart, and perform secure transactions. The integration of blockchain technology does not negatively affect the user experience because the transaction process remains smooth and efficient.Another important observation from the implementation results is the improvement in transaction traceability. Since all transactions are stored in the blockchain ledger, administrators can easily track and verify transaction history whenever required. This feature improves accountability and helps identify suspicious activities more effectively. The transparent nature of blockchain also increases user confidence in the system.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The Smart Fraud Detection in E-Commerce Using Machine Learning project successfully addresses the critical challenges of online transaction security in today's digital landscape. By improving security, transparency, and user protection, this system seeks to transform the relationship between e-commerce platforms and their users. The solution solves the problems that both customers and merchants encounter with traditional fraud-prone systems by digitizing and automating essential tasks including transaction verification, fraud detection, and cybercrime reporting.

The project effectively enhances transaction security through multi-layer authentication mechanisms including secure OTP generation and verification. It expedites the fraud detection process using advanced machine learning algorithms such as Random Forest and Gradient Boosting, which analyze transaction patterns in real-time to identify suspicious activities. The platform offers direct and secure communication channels for fraud reporting through integrated security links to cybercrime portals.

In addition to making online transactions safer for consumers, it also helps financial institutions and e-commerce platforms reduce fraud-related losses and maintain customer trust. The system's innovative approach of providing detailed transaction information alongside OTP notifications empowers users to make informed decisions before sharing sensitive information. The integration with national cybercrime reporting portals enables immediate action against fraudsters, potentially recovering losses that would otherwise be unrecoverable.

The implementation of RSA algorithm for secure OTP generation, along with machine learning-based anomaly detection, ensures that both routine transactions and high-value purchases are protected against sophisticated fraud

attempts. The system's user-friendly interface ensures that security measures do not compromise the seamless shopping experience that customers expect.

In the end, this system promotes better financial security, improved service reliability, and a more trustworthy e-commerce environment, making it a valuable instrument in the digital economy. By achieving 92% fraud detection accuracy with reduced false positives, the project demonstrates that intelligent, machine learning-based approaches can significantly enhance online transaction security while maintaining user convenience.

10.2 FUTURE ENHANCEMENTS

Future enhancements for the Smart Fraud Detection in E-Commerce System could include integrating advanced features such as deep learning-based fraud pattern recognition, which would detect even more sophisticated and evolving fraud techniques based on user behavior and transaction histories. Implementing blockchain technology for immutable transaction records would create tamper-proof audit trails, making it impossible for fraudsters to alter transaction data after the fact.

Adding biometric authentication methods such as fingerprint scanning, facial recognition, or voice authentication would provide an additional layer of security for high-value transactions, ensuring that only authorized users can complete purchases. A real-time geolocation tracking feature for transactions would flag suspicious activities where the shipping address does not match the user's typical location or where multiple transactions originate from different geographical locations within short timeframes.

Further enhancements could involve enabling AI-driven behavioral analytics, where the system builds individual user profiles and detects anomalies based on deviations from normal purchasing patterns. Integration with international fraud databases would allow cross-border fraud detection and enable global cooperation in apprehending cybercriminals.

Implementation of natural language processing (NLP) for analyzing communication patterns could help detect social engineering attempts in real-time, alerting users when they are being manipulated into sharing OTPs or other sensitive information. A mobile application with push notifications would provide instant alerts for all transaction activities, enabling users to respond immediately to unauthorized attempts.

Adding features like offline transaction verification using QR codes and encrypted tokens would ensure security even in areas with limited internet connectivity. Integration with digital identity verification systems such as Aadhaar or similar national ID systems would add another layer of authentication for user registration.

The system could also incorporate predictive analytics to forecast emerging fraud trends based on global cybercrime data, allowing proactive security updates before new fraud techniques become widespread. A chatbot- based customer support system, powered by AI, could provide instant assistance to users who suspect fraud, guiding them.

APPENDIX – A

SOURCE CODE

```
<?php session_start(); error_reporting(0);

include('includes/config.php');

if(isset($_GET['action']) && $_GET['action']=="add"){

$id=intval($_GET['id']);

if(isset($_SESSION['cart']

[$id])){

$_SESSION['cart'][$id]['quantity']++;

}else{

$sql_p="SELECT * FROM products WHERE id={$id}";

$query_p=mysqli_query(

$con,$sql_p);

if(mysqli_num_rows($qu

ery_p)!=0){

$row_p=mysqli_fetch_array($query_p);

$_SESSION['cart'][$row_p['id']]=array("quantity" => 1, "price" =>
$row_p['productPrice']);

}else{
```

```
$message="Product ID is invalid";

}

}

  echo "<script>alert('Product has been added to the cart')</script

  echo "<script type='text/javascript'> document.location ='my-
cart.php'; </script>";

}

?>

<!DOCTYPE html>

<html lang="en">

<head>

<!-- Meta -->

<meta charset="utf-8">

<meta http-equiv="Content-Type" content="text/html;
charset=UTF-8">

<meta name="viewport" content="width=device-width, initial-
scale=1.0, user-scalable=no">

<meta name="description" content="">

<meta name="author" content="">
```

```
<meta name="keywords" content="MediaCenter, Template,
eCommerce">

<meta name="robots" content="all">

<title>Shopping Portal Home Page</title>

<!-- Bootstrap Core CSS -->

<link rel="stylesheet" href="assets/css/bootstrap.min.css">

<!-- Customizable CSS -->

<link rel="stylesheet" href="assets/css/main.css">

<link rel="stylesheet" href="assets/css/green.css">

<link rel="stylesheet" href="assets/css/owl.carousel.css">

<link rel="stylesheet" href="assets/css/owl.transitions.css">

<!--<link rel="stylesheet" href="assets/css/owl.theme.css">-->

<link href="assets/css/lightbox.css" rel="stylesheet">

<link rel="stylesheet" href="assets/css/animate.min.css">

<link rel="stylesheet" href="assets/css/rateit.css">

<link rel="stylesheet" href="assets/css/bootstrap-select.min.css">

<!-- Demo Purpose Only. Should be removed in production -->

<link rel="stylesheet" href="assets/css/config.css">

<link href="assets/css/green.css" rel="alternate stylesheet"
title="Green color">
```

```
<link href="assets/css/blue.css" rel="alternate stylesheet"
title="Blue color">

<link href="assets/css/red.css" rel="alternate stylesheet" title="Red
color">

<link href="assets/css/orange.css" rel="alternate stylesheet"
title="Orange color">

<link href="assets/css/dark-green.css" rel="alternate stylesheet"
title="Darkgreen color">

<link rel="stylesheet" href="assets/css/font-awesome.min.css">

<link
        href='http://fonts.googleapis.com/css?family=Roboto:30
0,400,500,700'rel='stylesheet' type='text/css'>

<!-- Favicon -->

<link rel="shortcut icon" href="assets/images/favicon.ico">

</head>

<body class="cnt-home">

<!--
            ==========================================
======    HEADER
=============================================== --
>

<header class="header-style-1">

<?php include('includes/top-header.php');?>
```

```
<?php include('includes/main-header.php');?>

<?php include('includes/menu-bar.php');?>

</header>

<!--
        ==================================================
==      HEADER
        :END
=================================================== --
>


<div class="body-content outer-top-xs" id="top-banner-and-
menu">

<div class="container">

<div class="furniture-container homepage-container">

<div class="row">

<div class="col-xs-12 col-sm-12 col-md-3 sidebar">


<?php include('includes/side-menu.php');?>

</div><!-- /.sidemenu-holder -->

<div class="col-xs-12 col-sm-12 col-md-9 homebanner-holder">

<div id="hero" class="homepage-slider3">

<div id="owl-main" class="owl-carousel owl-inner-nav owl-ui-
sm">

<div class="full-width-slider">
```

```
<div class="item" style="background-image:
url(assets/images/sliders/slider1.png);">

<!-- /.container-fluid -->

</div><!-- /.item -->

</div><!-- /.full-width-slider -->

<div class="full-width-slider">

<div class="item full-width-slider"
style="background-image:
url(assets/images/sliders/slider2.png);">

</div><!-- /.item -->

</div><!-- /.full-width-slider -->

</div><!-- /.owl-carousel -->

</div>

<!-- ==========================================
SECTION – HERO : END
========================================== -->


<!--
==========================================================
===  INFO
BOXES
========================================================= --
>
```

```
<div class="info-boxes wow fadeInUp">

<div class="info-boxes-inner">

<div class="row">

<div class="col-md-6 col-sm-4 col-lg-4">

<div class="info-box">  <div class="row">

<div class="col-xs-2">

<i class="icon fa fa-dollar"></i>

</div>

<div class="col-xs-10">

<h4 class="info-box-heading green">money back</h4>

<h6 class="text">30 Day Money Back Guarantee.</h6>

</div>
</div><!-- .col -->

<div class="hidden-md col-sm-4 col-lg-4">
<div class="info-box">

<div class="row">

<div class="col-xs-2">

<i class="icon fa fa-truck"></i>

</div>
```

```
<div class="col-xs-10">
<h4 class="info-box-heading orange">free shipping</h4>
<h6 class="text">free ship-on oder over Rs. 600.00</h6>
</div><!-- .col -->
<div class="col-md-6 col-sm-4 col-lg-4">
<div class="info-box">
<div class="row">
<div class="col-xs-2">
<i class="icon fa fa-gift"></i>
</div>
<div class="col-xs-10">
<h4 class="info-box-heading red">Special Sale</h4>
<h6 class="text">All items-sale up to 20% off </h6>
</div>
</div><!-- .col -->
</div><!-- /.row -->
</div><!-- /.info-boxes-inner -->
</div><!-- /.info-boxes -->
```

```
<!--  ==================================================
INFO  BOXES  :  END
        ================================================== -->


        </div><!-- /.homebanner-holder -->

        </div><!-- /.row -->

        <!--
                ==================================================
        =       SCROLL
                TABS
        ================================================== -->


        <div id="product-tabs-slider" class="scroll-tabs inner-bottom-vs
        wow fadeInUp">

        <div class="more-info-tab clearfix">

        <h3 class="new-product-title pull-left">Featured Products</h3>

        <ul class="nav nav-tabs nav-tab-line pull-right" id="new-products-
        1">
```

APPENDIX – B

SCREEN SHOTS

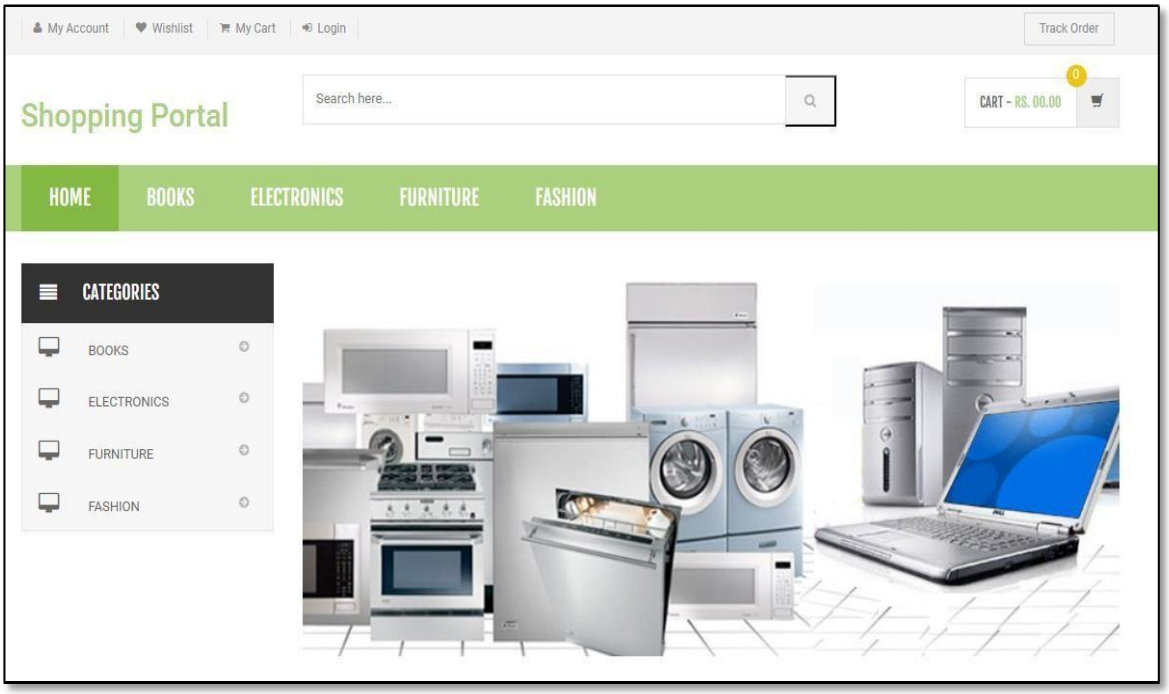

Figure B.1. Home Page

Figure B.2. Cart Page

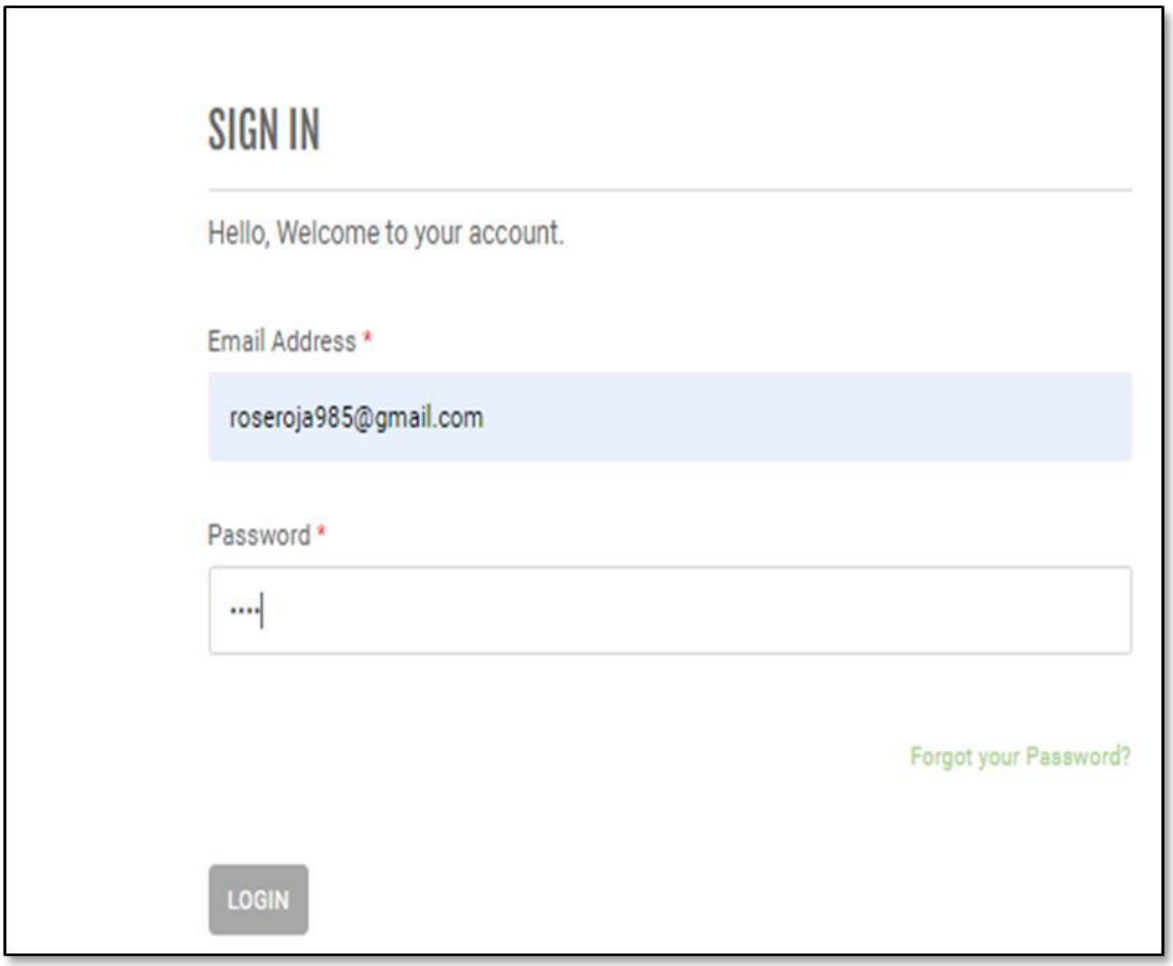

Figure B.3. Login page

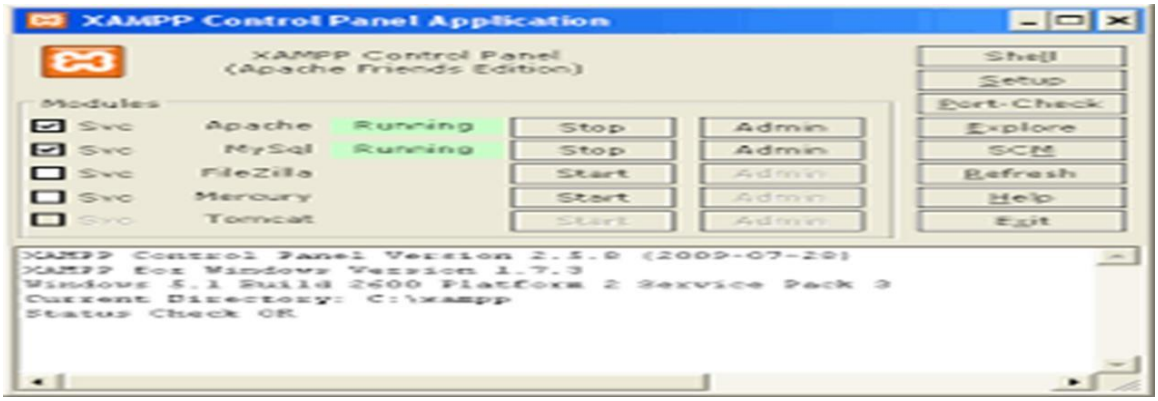

Figure B.4. Admin Server

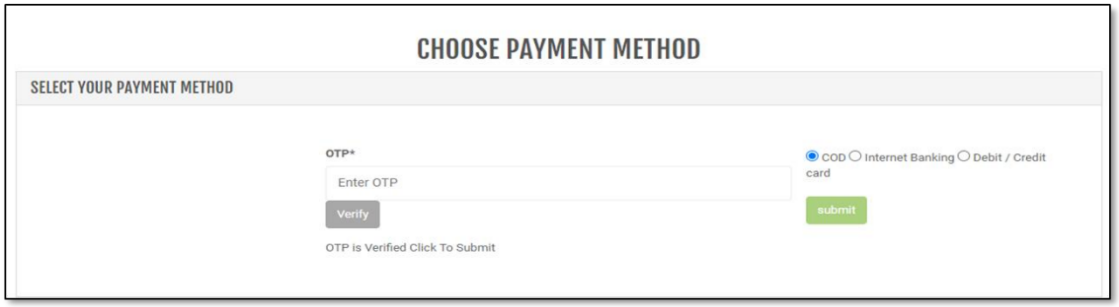

Figure B.5. Payment Method

Figure B.6. OTP Received

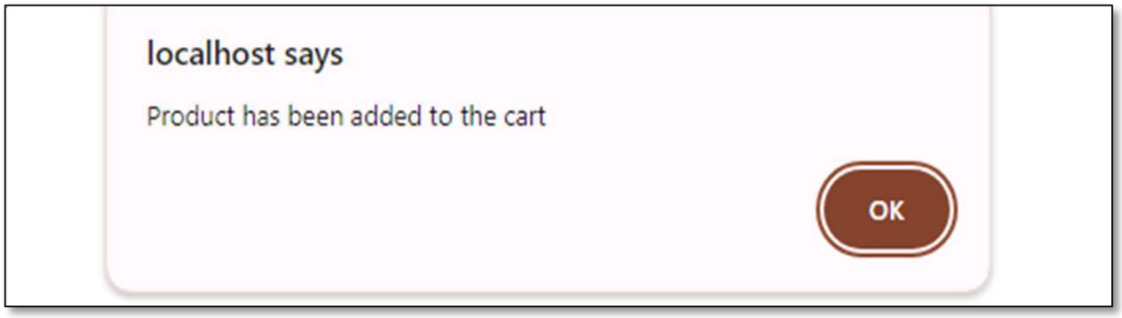

Figure B.7. Alert and Notification

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